1. Human Eye Structure & Working

* Cornea: Transparent outer cover, refraction of light.
* Iris & Pupil: Controls amount of light entering eye.
* Retina: Real & inverted image formation, rods & cones.

MIND MAP & IMPORTANT FORMULAS / CONCEPT SUMMARY

> Eye Lens focus adjustment: Accommodation power.

> Near Point: 25 cm, Far Point: Infinity for normal eye.

EXAM BOOSTER & PREVIOUS BOARD QUESTIONS (BSEB / CBSE)

Q1. Explain the main concept with suitable diagram and units.

Q2. Write short notes on key phenomena and practical applications.

Q3. Differentiate between primary principles and real-world examples.

Q4. Solve numerical questions based on standard Board exam paper patterns.

2. Defects of Vision & Corrections

* Myopia (Short-sightedness): Corrected using Concave lens.
* Hypermetropia (Long-sightedness): Corrected using Convex lens.
* Presbyopia: Old age defect, corrected using Bifocal lens.

MIND MAP & IMPORTANT FORMULAS / CONCEPT SUMMARY

> Causes of Myopia: Elongated eyeball or excessive focal length.

> Bifocal Lens: Upper part concave (far), lower convex (near).

EXAM BOOSTER & PREVIOUS BOARD QUESTIONS (BSEB / CBSE)

Q1. Explain the main concept with suitable diagram and units.

Q2. Write short notes on key phenomena and practical applications.

Q3. Differentiate between primary principles and real-world examples.

Q4. Solve numerical questions based on standard Board exam paper patterns.

3. Refraction through Glass Prism

* Prism disperses white light into 7 colors (VIBGYOR).
* Red light bends least, Violet light bends most.
* Angle of deviation depends on refractive index & wavelength.

MIND MAP & IMPORTANT FORMULAS / CONCEPT SUMMARY

> Spectrum: Band of 7 colored lights.

> Recombination: Upside down glass prism recombines spectrum to white light.

EXAM BOOSTER & PREVIOUS BOARD QUESTIONS (BSEB / CBSE)

Q1. Explain the main concept with suitable diagram and units.

Q2. Write short notes on key phenomena and practical applications.

Q3. Differentiate between primary principles and real-world examples.

Q4. Solve numerical questions based on standard Board exam paper patterns.

4. Rainbow Formation & Natural Phenomena

* Raindrops act as small glass prisms in atmosphere.
* Total Internal Reflection + Dispersion + Refraction = Rainbow.
* Observer must stand with back towards the Sun.

MIND MAP & IMPORTANT FORMULAS / CONCEPT SUMMARY

> Primary rainbow: Red outside, Violet inside.

> Atmospheric refraction causes twinkling of stars & early sunrise/late sunset.

EXAM BOOSTER & PREVIOUS BOARD QUESTIONS (BSEB / CBSE)

Q1. Explain the main concept with suitable diagram and units.

Q2. Write short notes on key phenomena and practical applications.

Q3. Differentiate between primary principles and real-world examples.

Q4. Solve numerical questions based on standard Board exam paper patterns.

5. Tyndall Effect & Scattering of Light

* Scattering of light by colloidal particles in a medium.
* Path of light beam becomes visible (e.g. sunlight through canopy).
* Scattered light intensity is inversely proportional to lambda^4 (Rayleigh scattering).

MIND MAP & IMPORTANT FORMULAS / CONCEPT SUMMARY

> Fine particles scatter blue light more strongly.

> Dust & smoke particles scatter light of all wavelengths.

EXAM BOOSTER & PREVIOUS BOARD QUESTIONS (BSEB / CBSE)

Q1. Explain the main concept with suitable diagram and units.

Q2. Write short notes on key phenomena and practical applications.

Q3. Differentiate between primary principles and real-world examples.

Q4. Solve numerical questions based on standard Board exam paper patterns.

6. Why Sky is Blue & Red Sunrise/Sunset

* Air molecules scatter shorter blue wavelengths more.
* At sunrise/sunset, light travels longer distance through atmosphere.
* Blue light scatters away, only longer red wavelengths reach observer.

MIND MAP & IMPORTANT FORMULAS / CONCEPT SUMMARY

> Danger signal lights are red because red scatters least by fog/smoke.

> In space (no atmosphere), sky appears dark black.

EXAM BOOSTER & PREVIOUS BOARD QUESTIONS (BSEB / CBSE)

Q1. Explain the main concept with suitable diagram and units.

Q2. Write short notes on key phenomena and practical applications.

Q3. Differentiate between primary principles and real-world examples.

Q4. Solve numerical questions based on standard Board exam paper patterns.

7. Key Diagrams & Ray Optics Summaries

* Ray diagram of Myopic eye & Concave lens correction.
* Ray diagram of Hypermetropic eye & Convex lens correction.
* Prism refraction diagram showing incident, refracted & emergent rays.

MIND MAP & IMPORTANT FORMULAS / CONCEPT SUMMARY

> Power of lens P = 1/f (in meters), unit: Dioptre (D).

> Concave lens power is negative (-), Convex lens power is positive (+).

EXAM BOOSTER & PREVIOUS BOARD QUESTIONS (BSEB / CBSE)

Q1. Explain the main concept with suitable diagram and units.

Q2. Write short notes on key phenomena and practical applications.

Q3. Differentiate between primary principles and real-world examples.

Q4. Solve numerical questions based on standard Board exam paper patterns.

8. Important Formulas & Board Revision

* Lens Formula: $1/v - 1/u = 1/f$
* Power $P = 1/f(m)$
* Angle of Prism A + Angle of Deviation D = i + e

MIND MAP & IMPORTANT FORMULAS / CONCEPT SUMMARY

> Power of accommodation allows eye lens to change focal length.

> Ciliary muscles contract for near vision, relax for far vision.

EXAM BOOSTER & PREVIOUS BOARD QUESTIONS (BSEB / CBSE)

Q1. Explain the main concept with suitable diagram and units.

Q2. Write short notes on key phenomena and practical applications.

Q3. Differentiate between primary principles and real-world examples.

Q4. Solve numerical questions based on standard Board exam paper patterns.

9. Eye Care & Safety Guidelines

* Keep reading distance 25-30 cm from eyes.
* Avoid reading in dim light or moving vehicles.
* Eat vitamin A rich foods (carrots, spinach, milk, papaya).

MIND MAP & IMPORTANT FORMULAS / CONCEPT SUMMARY

> Regular eye check-ups prevent permanent vision damage.

> Never rub eyes with dirty hands or direct water splashes.

EXAM BOOSTER & PREVIOUS BOARD QUESTIONS (BSEB / CBSE)

Q1. Explain the main concept with suitable diagram and units.

Q2. Write short notes on key phenomena and practical applications.

Q3. Differentiate between primary principles and real-world examples.

Q4. Solve numerical questions based on standard Board exam paper patterns.

10. One Page Quick Revision & Exam Booster

* Complete chapter summary with mind map & formulas.
* Assertion-Reason & Case-Based questions.
* Previous 5 Years Board Questions (BSEB Class 10).

MIND MAP & IMPORTANT FORMULAS / CONCEPT SUMMARY

> VIBGYOR mnemonic for spectrum colors.

> Defect correction summary table for quick review.

EXAM BOOSTER & PREVIOUS BOARD QUESTIONS (BSEB / CBSE)

Q1. Explain the main concept with suitable diagram and units.

Q2. Write short notes on key phenomena and practical applications.

Q3. Differentiate between primary principles and real-world examples.

Q4. Solve numerical questions based on standard Board exam paper patterns.